Closing Edge Loops at Quantization Time: Helix Constraints for T-Mesh Quantization

Research note — draft, not peer reviewed

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Abstract

Automatic quad remeshing often produces edge loops that wind round a part of the shape as helices instead of closing into rings. Bommes et al. [3] remove such *q-helices* from a finished mesh; we prevent them where the integer structure of the mesh is decided, in the quantization of a T-mesh. A straight walk across the patches of a T-mesh that returns to its first side, a *turn*, has a *shift* that is an integer linear form in the arc lengths; at whole lengths it is how far the strip of quads following the turn comes back beside itself, and every strip of the mesh is such a walk. We add the constraints that close turns to a minimum-deviation quantizer [13], through a least-change projection, repair walks and rounds of constraint generation on the strips of the mesh, and keep the candidate with the fewest spiral edges, counted by a test that tells helices from closed, open and border loops. Over twelve models at three budgets near 3,000 quads, the engine leaves 4.65% of the edges on spiral loops (16.7% without closing, in single runs of an earlier engine); on the seven models that two other open-source remeshers also mesh validly, 1.8% against 38.8% and 21.7%; on a human body, single runs give 1.6%, against 2.5% for the hand-made mesh. An exact analysis of the closing system of that body shows that the conflict between closing turns and keeping singularities apart is real but local, that the heuristic rather than the constraints leaves most turns shifted, and that closing more turns does not by itself make fewer spirals: the strips move with the lengths, and the choice of the turns to close decides.

1 Introduction

Field-aligned quad remeshing has reached a level where element shape and alignment are rarely the limiting factor [5, 6]. What remains is structure: the way edge loops run across the surface. Artists expect a loop round an arm, a leg or a neck to be a ring. Automatic methods often produce a helix instead: the loop goes round, comes back one or a few rows further, and winds along the whole part (Figure 3). Bommes et al. showed that helical configurations are a main reason why the base complex of an otherwise good quad mesh stays dense [3], and removed them from the finished mesh.

Whether a loop closes is decided upstream, when a quantizer assigns whole lengths to the arcs of a T-mesh traced in a seamless parametrization [8, 13, 23]. Quantizers choose these lengths for consistency, validity and closeness to the parametrization; the loops close or spiral

*This note is distributed under the Creative Commons Attribution 4.0 International licence (CC BY 4.0). The code of openclare is under the GPL-3.0-or-later of the repository. The note describes the engine at commit `9a2851905a25` of `https://github.com/marcgardent/openclare`. A French translation is in `paper.fr.pdf`, beside this note.

as a side effect. This paper constrains the loops in the quantization itself. Section 3 defines *turns*, straight walks across the patches of a T-mesh that return to their first side, and shows that the *shift* of a turn is an integer linear form in the arc lengths that gives, at whole lengths, the pitch of the strip of quads following it, and that every strip of the mesh is such a walk. Section 4 adds the closing of turns to a minimum-deviation quantizer through a least-change projection, repair walks and rounds of constraint generation on the strips, with safeguards on folds, distortion and budget. Section 5 gives a share of spiral edges that tells helices of pitch up to four from closed, open and border loops; loop counts and loop simplicity scores exist [18, 22], but they do not separate helices from open loops. Section 6 measures the method in the open-source remesher openclare against two open-source remeshers and a mesh modelled by hand, and Section 7 solves the closing system of a human body exactly to see where the heuristic stands: the conflict between closing turns and keeping singularities apart is local, the heuristic leaves most of the shifted turns that no constraint forces, and yet closing more turns does not by itself reduce the spirals, because the strips of the mesh move with the lengths. Section 8 states the limitations and the open problems.

This work began as a programming project, openclare, an open-source quad remesher, and became research when engineering alone could no longer tell which results could be guaranteed and which computations could be saved. Each result used here answers such a question. The shift of a turn is linear in the arc lengths, so closing a loop becomes a linear constraint of the quantizer rather than a test after the fact; its parity follows the length of the turn, so some repairs are known to be impossible before they are tried; and choosing the largest set of turns to close is NP-hard, which is why the engine keeps to a heuristic, with an exact solver only on demand.

The theory behind the conflicts, which reads the shift as translational holonomy of the flat metric and derives obstructions to closing from the singularities, a lower bound on the turns every quantization leaves shifted, and the NP-hardness of choosing the turns to close, is the subject of a companion paper [12]; we state its results in one sentence where we need them, without proof. Statements here are *proved* when they come with a proof, *measured* when they come from exact arithmetic or solver output on the systems openclare exports, and *conjectured* otherwise.

2 Background and related work

T-mesh quantization. Mixed-integer quadrangulation [2] and integer-grid maps [4] decide the integer structure of a quad mesh in their quantization. Campen et al. quantize via the integer lengths of the arcs of a T-mesh traced in a seamless parametrization [8]. Lyon et al. solve a constrained version as an integer linear program, bounding the directional deviation of the separatrices of the resulting layout, and require, as openclare does, that the quantization keep singularities apart [23]. Heistermann et al. cast quantization as a minimum-deviation flow on a bidirected graph and give a fast approximate solver and an exact refinement [13]; openclare follows their formulation. To our knowledge none of these constrains the helices.

Helices after meshing. Bommes et al. [3] define a *q-helix* as a dual parametric line with an integer *pitch* h and *turn length* τ: walking τ steps along it is equivalent to h side-steps. They detect minimal q-helices by the self-intersections of orthogonal dual lines and remove them greedily with grid-preserving operators, closed dual paths of shift and collapse steps that keep the mesh all-quad and its singularities in place, followed by tangential smoothing; their experiments remove most helices (from 17–52 down to 1–23 per model). They note that choosing the pairs of singularities to align directly in the parametrization, as Myles et al. do for T-meshes [24], is often complicated or even ambiguous to automate. Razafindrazaka and Polthier [28] replace the base complex of a finished mesh by an optimal one, found as a perfect matching on a graph

over its singular vertices, and note that the operators of [3] leave some spirals. Both work after meshing, not in the quantization.

Periods round a tube. QuadCover [16] lifts the cross field to a branched covering where it becomes a vector field. The stripe parameterization of tubular surfaces [17], built on it, measures the jump ("gap") of the parameter across each cut path and rounds it to the nearest whole number; the gap across a loop round a tube is what we call the shift of a turn, and rounded to a non-zero whole number it gives a helix of that pitch. That paper treats spiralling stripes as a legitimate outcome, and notes that rounding the gaps for least distortion is a hard combinatorial problem that it does not solve exactly.

Flat metrics and finite trajectories. Declaring every quad a unit square turns a quad mesh into a flat metric with cone points at its irregular vertices. Chen et al. [9] characterize the metrics that arise this way; one of their conditions is a *finite geodesic lamination*, the straight lines of the field being finite loops, which they assume rather than construct. Lei et al. [20] show that a closed quad mesh induces a meromorphic quartic differential and list as an open problem the guarantee that all its trajectories are finite. A trajectory that is not finite is precisely an edge loop that does not close. A quadratic differential all of whose non-critical trajectories close is a Jenkins–Strebel differential [30]; the complement of its critical graph is a union of flat cylinders, and a helix is a cylinder glued back with a shift of a whole number of rows. On a closed surface the valences $k(v)$ of a quad mesh satisfy $\sum_v (4 - k(v)) = 4\chi$: with valences three and five only, a closed human body needs at least eight vertices of valence three, but a hand-made base mesh has hundreds, and their count says nothing of where the loops go.

Loops by construction. Dual loops meshing [7] builds a layout from loops traced on the surface, which close by construction; Lei et al. [19] build quad meshes from a Strebel differential with prescribed cylinders, and skeleton-guided layouts [32] close the loops round limbs by design. Myles et al. [25] trace the field into a partition into patches, Pietroni et al. [27] select traced separatrices by a binary linear program, and Razafindrazaka et al. [29] obtain a layout as a minimum-weight perfect matching. These methods build coarse layouts, and none starts from a given cross field and singularity configuration to make a dense mesh whose loops close.

Measuring loops. QuadGPT [22] counts *quad rings* (closed loops of faces) and *quad lines* (open strips) and uses them as reward signals for preference optimization, without telling spirals from open lines. SQuadGen [18] scores the loops of a layout by their self-intersections and their rotation index, and reports the area share of simple loops; it does not tell helices from open loops either.

3 Turns of a T-mesh

A T-mesh is a partition of the surface into rectangles, the *patches*, whose sides are made of *arcs*; nodes where a trace stops on another are T-junctions, and the singular vertices of the field are nodes. Each patch P has local coordinates with its first corner at the origin and its sides along the axes. A quantization assigns every arc a a length $\ell_a \in \mathbb{Z}_{\geq 0}$ such that opposite sides of every patch have equal length, the *balances* $B\ell = 0$; the quad mesh is then read off as the integer grid of each patch.

Definition 1 (Turn)**.** *A* turn γ *is a straight walk across patches, from an offset of its first side that meets no node: it enters a patch through a side, leaves through the opposite side at the same distance from the corresponding corner, enters the neighbouring patch across the arc it reaches, and so on, until it comes back into its first patch through its first side.*

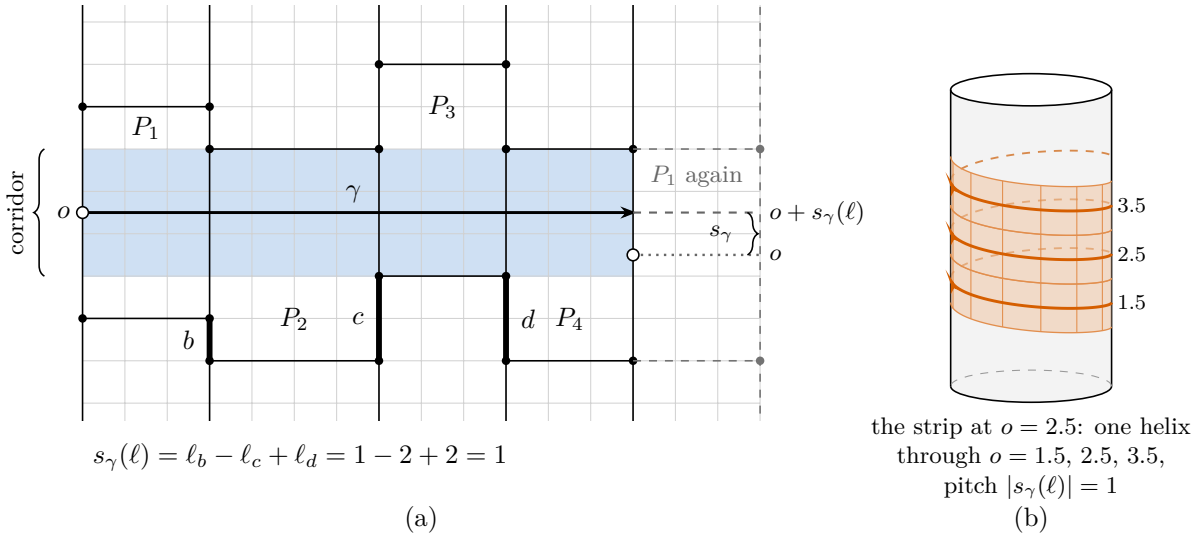


Figure 1: A turn and the helix it makes. (a) A band of a T-mesh on a cylinder, cut open along the first side of P_1, which is drawn again, dashed, past P_4, where it is glued; every node is at a whole position and the faint grid is the quad mesh at these lengths. The turn γ leaves the first side at offset o and crosses P_1 to P_4; measured from the first corner of each patch, its offset grows by ℓ_b, drops by ℓ_c and grows by ℓ_d, so it comes back at $o + s_\gamma(\ell)$ with $s_\gamma(\ell) = \ell_b - \ell_c + \ell_d = 1$. The corridor (shaded), the offsets whose walk crosses the same arcs, is $1 < o < 4$. (b) The strip of quads at $o = 2.5$ on the cylinder: each round comes back one row beside itself, through $o = 1.5$, 2.5 and 3.5, then leaves the corridor: a helix of pitch 1 (Proposition 1).

Crossing from patch P to patch Q across an arc composes a quarter-turn rotation of the local frame with a translation whose components are signed sums of arc lengths along the rims of P and Q, the position of a rim point being the sum of the lengths of the arcs between it and the first corner.

Proposition 1 (The shift of a turn is linear)**.** *For a turn γ there are integer coefficients c_a, independent of the lengths, such that after a full turn a point of the first side has moved across the walking direction by*

$$s_\gamma(\ell) = \sum_a c_a \, \ell_a .$$

For whole lengths ℓ, let o be a half-integer offset of the first side whose walk at the lengths ℓ crosses the same patches as γ (the set of such offsets is the corridor *of γ at ℓ). Then the dual line of the quad mesh at o comes back at $o + s_\gamma(\ell)$: it closes after one round if and only if $s_\gamma(\ell) = 0$. If moreover $o + j\, s_\gamma(\ell)$ lies in the corridor for $j = 1, \dots, t-1$, it winds t times round as a helix of pitch $|s_\gamma(\ell)|$ in the sense of [3].*

Sketch. Each crossing is a map $x \mapsto R^k x + t$ with R the quarter-turn and t a vector whose entries are sums of $\pm\ell_a$. The walk re-enters its first side inward, so the composition along γ has trivial rotation and is a translation $T_\gamma(\ell)$, linear in ℓ with integer coefficients. Its component along the walking direction does not move a point of the first side off that side; its component across is $s_\gamma(\ell)$. Singularities are nodes, never inside a patch, so the quad mesh is a regular grid along the walk, and a dual line at distance o from the first corner, as long as it crosses the same patches as γ, comes back at distance $o + s_\gamma(\ell)$. □

The component of $T_\gamma(\ell)$ along the walking direction is the *length* $\alpha_\gamma(\ell)$ of the turn, also an integer linear form, which at whole lengths is the number of quads of the strip in one turn. The

shift is the across component of the translational holonomy of the flat metric along the turn, linear in the lengths as any period of a flat cone metric [31, 33]; for a ring of quads it is the twist of a cylinder. What is specific here is its expression in the arc lengths of a T-mesh, as a constraint of the quantization. The companion paper [12] shows that it is set by the singularities: for a turn whose curve is simple, it equals up to sign the across component of the Burgers vector of the singular vertices on one side, so that two parallel turns bounding an annulus free of singular vertices have the same shift, and a helix is a property of a whole cylinder of turns, not of one of them. In openclare (`quantization/tubes.rs`) the shift is built once per turn as an expression in the arc lengths, by composing the rim expressions of the crossed patches; evaluated at the real lengths of the parametrization it gives the *real shift*, the twist the field itself gives the cylinder. Figure 1 shows a turn, its corridor and the helix its shift makes.

The closing constraint $s_\gamma(\ell) = 0$ thus speaks of a fixed sequence of crossed patches, and describes a strip of the mesh only together with the corridor. Explicitly, let γ cross the arcs $a_1, \dots, a_m$ in turn; the walk from offset o crosses the side holding a_j at the distance $\epsilon_j o + q_j(\ell)$ from its first corner, with $\epsilon_j = \pm 1$ and q_j an integer linear form, and a_j starts at the distance $e_j(\ell)$, a sum of arc lengths. At whole lengths and for a half-integer o, the corridor is cut out by the inequalities

$$e_j(\ell) + \tfrac{1}{2} \;\le\; \epsilon_j\, o + q_j(\ell) \;\le\; e_j(\ell) + \ell_{a_j} - \tfrac{1}{2}, \qquad j = 1, \dots, m, \tag{1}$$

two per crossing. The closing steps of Section 4 impose the shifts but not these inequalities; Section 7 measures what this costs. The strips themselves are all walks of this kind.

Definition 2 (Strip)**.** *For whole lengths ℓ balancing every patch, with every patch side positive and no two singular vertices at the same point, the quad mesh is, in each patch, the integer grid of its sides. A* strip *is a maximal sequence of quads, each sharing with the next the edge opposite the one it shares with the previous.*

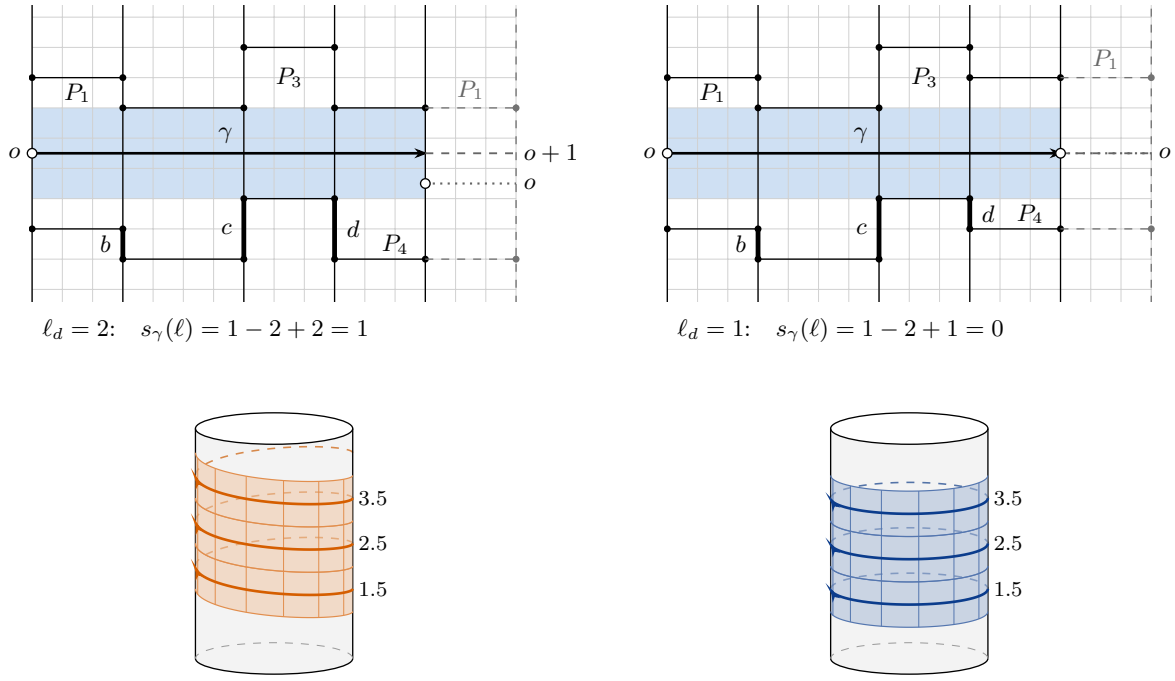
Proposition 2 (Completeness at given lengths)**.** *Let ℓ be as in the definition. Every strip of the quad mesh that crosses a side σ of a patch P at the quad edge $[k, k+1]$ of σ is the walk taken* with the lengths ℓ *from σ at offset $k + \frac{1}{2}$. On a closed surface the strips are thus the cycles of the walk on the half-integer stations of all sides, each oriented strip one cycle, and a strip closes after one round exactly when the turn it follows has $s_\gamma(\ell) = 0$. With borders, the strips that reach a border are open paths of the walk.*

Proof. Inside P, the quads of the strip are the unit squares between the grid lines at k and $k+1$ across σ; their dual line is the straight line at $k + \frac{1}{2}$, which reaches the opposite side at the same distance from the corresponding corner, the move of the walk. The crossing map to the neighbouring patch is a quarter-turn followed by a translation with whole components: it maps the grid of P onto the grid of the neighbour and the midpoint of a quad edge onto the midpoint of a quad edge. Every node lies at a whole distance along the sides, so a walk at a half-integer offset never meets a node and the arc it crosses is determined; an arc of length zero holds no half-integer and no walk crosses it. Strip and walk thus continue together, patch after patch; positive sides make every station lie on a quad edge of the mesh. □

A walk from a fixed share of the real side lengths samples the turns; Proposition 2 says where the turns of a given mesh are *all* found: at the half-integers of its whole lengths. The constraint rounds of Section 4 use it.

4 Closing turns in quantization

This section describes the engine at commit `9a2851905a25`, in `crates/openclare/src/quantization/`. The quantizer minimizes a separable convex cost of the whole lengths, in the



(a) before: $s_\gamma(\ell) = 1$

(b) after: $s_\gamma(\ell) = 0$

Figure 2: The turn of Figure 1 before and after closing, on the same T-mesh. (a) With $\ell_d = 2$, $s_\gamma(\ell) = 1$: the strips of the corridor form one helix. (b) The column of P_4 sits one unit higher: ℓ_d and the arc above P_4 on the same side are one shorter, no node passes another, and only lengths have changed. Now $s_\gamma(\ell) = 1 - 2 + 1 = 0$, and the corridor holds three closed rings, at $o = 1.5$, 2.5 and 3.5.

minimum-deviation-flow formulation of [13], subject to the balances, by cancelling improving walks in a strip network (a simpler solver than their matching-based one). Only the structure is theirs: their objective is a sum of absolute deviations, ours is quadratic. The cost of arc a is $w_a(\ell_a - \lambda r_a)^2$ with $w_a = 1/(\lambda r_a + 1)$, plus a term that draws a vertex onto each fold of the surface the arc crosses and a penalty for going below a lower bound. The real lengths r_a carry a fixed per-arc perturbation of at most 0.5%, so that arcs of equal length do not all change at the same scale; the scale λ is searched for the number of quads asked for, and arcs whose zero length would break the structure of the mesh, by bringing two singular vertices together, are held at length at least one, the *bounds of separation*. Closing turns is a side constraint that is not a flow constraint; Figure 2 shows its effect on one turn.

Finding turns. From points at given shares along each side of each patch, we walk straight across patches and keep the walks that return. Not every shifted turn should close: a turn round a part of the shape whose real shift is far from zero is not a rounding artefact, the field winds round the part, and a turn round a group of singularities may legitimately come back shifted. A turn is therefore *held* if its real shift is above 0.05, its *lean*, the real shift as a share of its length, below a bound, and its length below a multiple of the side of the T-mesh (the side of a square of its area). Two turns with the same shift expression up to sign are one ring and held once. Eight *ways* are tried: starting shares $\{0.5\}$ or $\{0.2, 0.5, 0.8\}$, lean bounds 0.05 or 0.12, length bounds 2 or 6 sides; a way that holds the same turns as another is skipped.

Least-change projection. The real lengths r_a, at scale one, are changed as little as possible, in the weighted least-squares sense, for every held turn $\gamma \in H$ to have zero shift and every patch to stay balanced:

$$\min_x \sum_a v_a \,(x_a - r_a)^2 \quad \text{s.t.} \quad Bx = 0, \quad s_\gamma(x) = 0 \;\; \forall \gamma \in H, \qquad v_a = \frac{1}{r_a + 1},$$

solved with a sparse Cholesky factorization of the balances, computed once per quantization, and a small dense Schur complement for the turns. Arcs the solution would make negative are held at zero and the system solved again (an active set). If the system cannot be met, the held turn left with the largest shift is let go and the projection repeated. The projected lengths x then take the place of r in the cost above, and λ is searched on them.

Integer search and repair. The integer search of the quantizer runs on the projected targets. Rounding leaves some turns shifted by one or a few rows. Repair walks are then taken one at a time: a walk is guided by a bias along the arcs of the shifted turns, and accepted if it reduces the total shift, costs little in deviation, keeps the structure intact and the number of quads within tolerance. A second, *aimed* repair treats each held turn still shifted on its own: the bias is put on its arcs alone, and a walk is accepted if it shifts that turn less and the others no more. Two results of the companion paper [12] prune these walks. On a closed sphere the shift of every turn has, at all whole balanced lengths, the parity of its length, $\alpha_\gamma(\ell) \equiv s_\gamma(\ell) \pmod 2$, so a walk that keeps the length of a turn of odd shift cannot close it; on the pieces that are closed spheres such walks are turned down without being judged. And a turn whose shift equals, modulo the balances, a non-negative combination of arc lengths is never negative and vanishes only when those arcs do; such a turn already at its least is not aimed at. The debug builds of the engine check the integer coefficients of the shifts and the parity.

Rounds of constraint generation. After a way whose lengths are admissible (within the budget and the deviation cap below), up to three rounds of lazy constraint generation follow; the constraints added are equalities on the shifts, not integrality cuts. Every strip of the whole lengths is walked (Proposition 2; up to 6 times the side of the mesh); those that come back shifted although their real lengths bring them back within 0.12 of their length are held as well, and the projection and the integer search are run again from the lengths of the round before. In these rounds the projection does not use the active set and the repair is not aimed. Each round gives one more candidate.

Selection and safeguards. For every candidate, the quad mesh connectivity is built and its share of edges on spiral loops measured (Section 5). A candidate is kept if it spirals less than the plain search (without closing), its number of quads is within the tolerance (or no further from the budget than that of the plain search), its mean absolute deviation from the real lengths is at most $\max(1.5\,d_0,\ d_0 + 0.2)$ with d_0 that of the plain search, and it leaves the vertices no further from the folds of the surface than the plain search did, by more than chance would. The three with the fewest spirals are kept, in that order, then the plain lengths. The ways are independent and run in parallel. After meshing, if the first candidate makes a face fold over the surface, the others are meshed in turn until one does not, and the mesh with the fewest folds, then the fewest spirals, is kept.

5 Measuring loops

An edge loop is followed straight through every regular vertex (valence four, interior), each edge belonging to one loop. A loop is *closed* if it comes back to its start; *border* if it runs from border to border; *open* if it stops at an irregular vertex or a non-quad face; and *spiral*, whatever its ends,

	spiral edges	Hausdorff (‰)	corner deviation (deg)
no closing	16.7%	8.77	4.75
closing, one way kept by spirals	9.7%	8.57	4.85
+ leaning ways, aimed repair	9.1%	8.58	4.91
+ longer turns, deviation cap	6.0%	8.68	5.01

Table 1: Parts of the method: mean over the twelve models at 3,000 quads, one run per model, engines of commits `7dc1727`, `c86740f` and `cf4a82c`. Each line adds to the one above only roughly: between the second and third lines the baseline moved (commit `ab8d89c`, 10.1% to 9.5%). The constraint rounds, the active set and the floor of the cap are not in the table; the engine at the cited commit gives 4.65% over three budgets.

if a strip of quads leaving it across meets it again, the same way round, within $k_{\max} = 4$ quads. The last test catches helices of pitch up to four, including bundles of interleaved helices [3], while a ring is met again only round a whole torus or over a cap the other way round. We report the share of edges on each kind of loop, edges of borders left out. The same count chooses among the candidates of Section 4 and colours the spiral loops in the figures.

6 Experiments

All numbers come from the openclare bench, measured on the output meshes against their sources; Appendix A gives the engine version and the source of every figure. The engine is deterministic: one run per model and budget. Models: twelve organic and mechanical models for development, and a corpus of 33 models for the full bench, among them two human bodies from Blender's Essentials library (CC0) whose base meshes, modelled by hand, serve as references. Times were measured on one machine, an Intel Core i7-7700K (4 cores, 8 threads, 4.2 GHz, 2017) with 62 GB of memory, shared with other jobs whose load was not recorded: they are wall-clock times and give orders of magnitude only. Linear and mixed-integer programs were solved with HiGHS 1.8.0 [15] through SciPy 1.16.3 [34] (`linprog` and `milp`), with default options except the time limits stated.

Spirals on the twelve development models. The share of edges on spiral loops is chaotic on a single model: a change of one per cent of the budget moves it as much as a real change of the method. We therefore judge the engine by its mean over the twelve models at three budgets, 2,970, 3,000 and 3,030 quads (36 runs). At the cited commit the engine leaves 4.65% of the edges on spiral loops (2.89%, 4.97% and 6.09% at the three budgets; 102 spiral loops in the 36 meshes, all valid), with 36.7% of the edges on closed loops, a mean distance to the source of 0.377‰ of the diagonal and a mean corner deviation of 4.90 degrees. Table 1 shows the contribution of the parts of the method, measured with the engine versions that introduced them, in single runs at 3,000 quads: from 16.7% of spiral edges without closing to 6.0%. Over these versions the Hausdorff distance stays within about 2% and the mean corner deviation within about 6%; the longer turns and the deviation cap trade a little of both for a third fewer spiral edges.

Against other remeshers. The last full bench committed (commit `83f0967`, 3,000 quads; one run per model for openclare and AutoRemesher [11], the median of five seeds for QuadriFlow [14], run through Blender) used an engine a few changes away from the cited one, which give 4.54% instead of 4.65% on the twelve models at three budgets. The tools do not mesh the same models: QuadriFlow produces a mesh for 22 of the 33 models, AutoRemesher for 18. On the 7 models that all three mesh validly in every run, the mean share of spiral edges is 1.8% for openclare, 38.8% for QuadriFlow and 21.7% for AutoRemesher; on the 15 models where all

three produce a mesh, valid or not, 4.2%, 27.2% and 27.3%. Without closing (commit `4e9620c`) openclare had 16.6% and 15.5% on the same subsets.

Against the hand-made mesh. The male body of the Essentials library has a base mesh of 10,590 quads modelled by hand, with 2.5% of spiral edges in 3 spiral loops and 31.6% of its edges on closed loops. Remeshed by openclare at that count with the closing of Section 4, single runs of the engines of commits `b0d37fa` to `8fd1932` give 1.6% of spiral edges, in 9 spiral loops: below the hand-made mesh, but by a few loops only. One such run has 33.2% of its edges on closed loops, a mean distance to the source of 0.19‰ of the diagonal and a mean corner deviation of 4.1 degrees, against 0.17‰ and 9.2 degrees for the hand-made mesh; these fidelity figures compare meshes made for different ends, the hand-made one being a base mesh whose distance and angles are choices of its modeller. The lean bound matters on this body: with the lean bound 0.05 alone (commit `ab8d89c`), one leg was a single helix from hip to ankle, its turns leaning more than that bound allows, and the mesh had 12.1% of spiral edges.

Figure 3 shows the body meshed by openclare with and without closing, next to the hand-made mesh and QuadriFlow's, with a later engine than the runs above (commit `89232b0`; its numbers are in the caption). Without closing, helices wind round the torso, both legs and the neck. With closing, the loops round the limbs and the torso are rings, as on the hand-made mesh, and the spiral loops left are short ones at the mouth, the forehead and the fingers. QuadriFlow winds helices round the torso, the shoulders, the legs and one arm. Closing is not better everywhere (Figure 4): at the left wrist it leaves spiral loops on the thumb and the back of the hand where the mesh without closing has closed ones, and near the right eye both openclare meshes keep short spiral loops, while closing turns a loop across the forehead into a ring.

7 An exact look at the closing system

How far is the heuristic of Section 4 from what the constraints allow? An experimental hook of openclare (branch `rd-exact`, on an engine without the constraint rounds and the floor of the deviation cap, that is before commit `3b3122d`) exports, for each way of finding turns, the system it works on, for the male body at 10,590 quads. Its T-mesh has 1,295 nodes (1,034 T-junctions and 261 singular vertices: 133 of valence three, 127 of valence five and one of valence two), 2,069 arcs and 776 patches, hence 1,552 balance equations, one per patch and direction. The export gives the real lengths, every turn found with its shift (on all 1,299 turns, composing the crossing maps along the walk gives the exported shift), and the lower bounds the integer search placed on arcs to keep singularities apart (28 to 36 arcs held at length one). These bounds are the engine's choice of *which* arc of a chain between two singular vertices carries their separation: validity asks a chain to keep a positive total length, not a given arc. Every statement below is relative to them. We call P_{eng} the balanced lengths that meet them; it is neither a subset nor a superset of the valid quantizations, since it excludes valid lengths that put the separation on another arc of the chain and allows lengths that bring other arcs to zero. The ways are numbered as exported: way $m + 2j + 4i$ starts from the shares $\{0.5\}$ ($m = 0$) or $\{0.2, 0.5, 0.8\}$ ($m = 1$), with the lean bound 0.05 ($j = 0$) or 0.12 ($j = 1$) and the length bound 2 ($i = 0$) or 6 ($i = 1$) sides. Way 2 is the one the engine keeps for this body.

The conflict is real but local. All held turns can be closed together if the lengths are only asked to be non-negative; with the bounds of separation, they cannot. By Farkas' lemma a set of turns cannot be closed within the bounds exactly when some combination of their shifts equals, modulo the balances, a non-negative combination of arc lengths that charges an arc held open, and the companion paper [12] reads these certificates on the singularities. In four ways (0, 1, 2 and 4) one held turn cannot even be closed on its own: the same turn round the left wrist, of length 7.0, whose shift equals, modulo the balances, $-(\ell_{1921} + \ell_{1937})$, where arc 1937, 0.08 quad

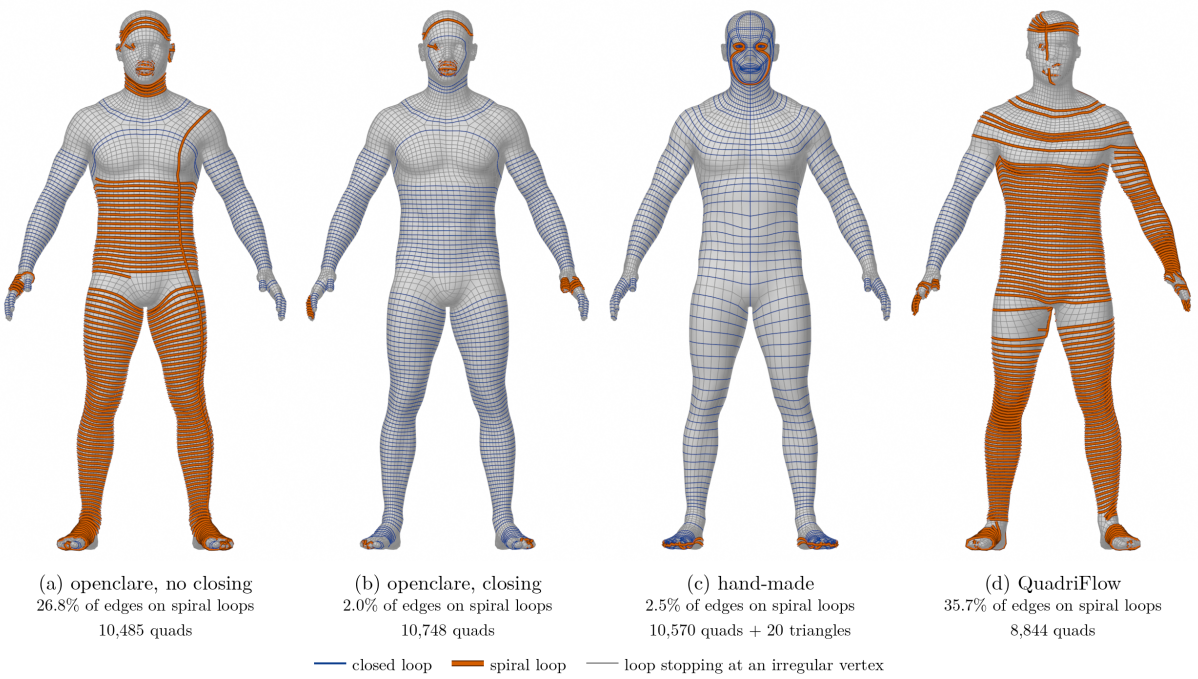


Figure 3: The male body of the Essentials library, 10,590 quads asked for: (a) openclare without closing, (b) openclare with the closing of Section 4, (c) the base mesh modelled by hand, (d) QuadriFlow run through Blender. Edge loops are coloured by their kind in the measure of Section 5: closed loops dark blue, spiral loops vermillion (wider, with a dark rim), loops that stop at an irregular vertex grey. Meshes (a) and (b) come from the engine at commit `89232b0`, one deterministic run each, not from the runs quoted in the text (Appendix A). Measured on the whole meshes, they have 10,485, 10,748, 10,570 (plus 20 triangles) and 8,844 quads, 26.8% (18 loops), 1.95% (15), 2.55% (3) and 35.7% (43) of their edges on spiral loops, which the panels round to one decimal, and 10.2%, 32.6%, 31.6% and 2.7% on closed loops. QuadriFlow's mesh has 3 inverted faces and a Hausdorff distance of 31‰ of the diagonal.

edges long in the parametrization, is held at length at least one (the certificate checked in exact rational arithmetic for way 0). Whether $\ell_{1937} = 0$ would merge two singular vertices, that is whether every valid quantization meets this bound, we did not check. Giving up the least total amount of the bounds needed to close all held turns, a linear program, shows that in ways 0, 2 and 4 the bound of that single arc stands in the way. The other conflicts gather in clusters of singularities at the eyes, the hands and the feet.

How far the projection is. With every bound kept, the largest set of held turns that can be closed together is found by a mixed-integer program with one binary per turn (Table 2). The projection of the engine of the export keeps far fewer: it removed, one at a time, the turn left most shifted by a least-squares projection whose negative lengths were clamped to zero, and a single clamped arc could make it drop dozens of turns that conflict with nothing. The engine at the cited commit holds such arcs at zero instead (the active set of Section 4). In whole numbers, the same mixed-integer program did not finish within fifteen minutes on the two systems tried.

The rounding ignores the shifts. Feeding the integer search of the engine better targets shows what survives it (Table 3). Aiming at the lengths of the linear program closes fewer loops in the end: its weighted L^1 objective puts the changes on few arcs, the deviation from the real lengths grows, and the variants with the fewest spirals are refused by the deviation cap. The active-set projection keeps every held turn closed in real numbers, yet after the integer search

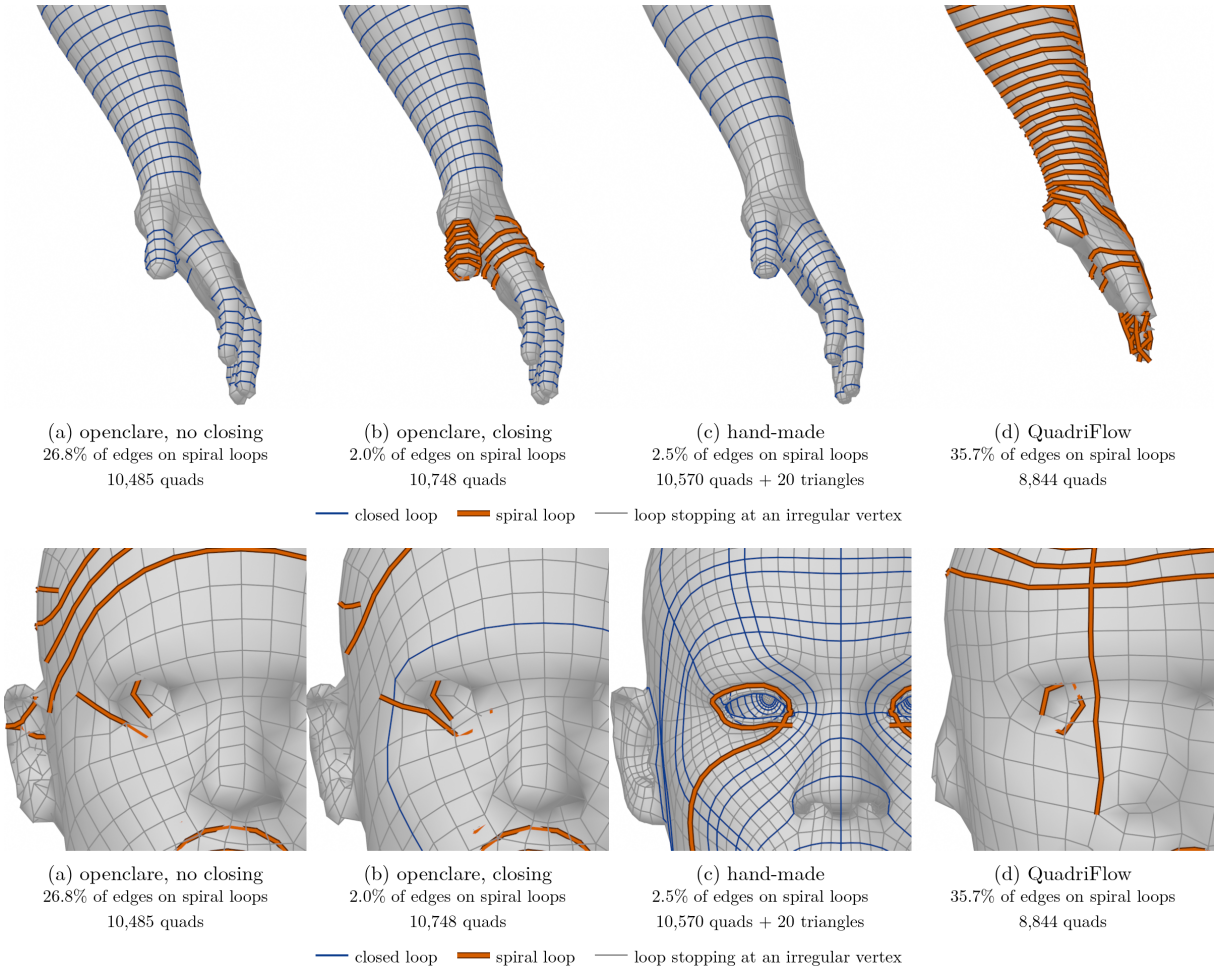


Figure 4: Close-ups of the four meshes of Figure 3, in the same order and colours: the left wrist and hand (top; the body's left, on the viewer's right in Figure 3) and the right eye (bottom). At the wrist closing does not help locally: mesh (b) has spiral loops on the thumb and the back of the hand where mesh (a) has closed ones, although over the whole body closing removes most spiral edges. At the eye, closing turns a loop across the forehead into a ring, and short spiral loops remain near the eye on both openclare meshes; the loop round the eye of the hand-made mesh is itself counted as spiral. In the T-mesh exported in Section 7, from an earlier engine, the left wrist holds the turn that in four of the eight ways no lengths within the bounds of separation can close even on its own.

117 of 169 are shifted again and the mesh spirals more; with the clamped projection, too, about 40% of the turns it kept are shifted after the integer search. The integer search follows the targets but not the shifts.

Whole lengths are not what is missing. The balances are the incidence matrix of a bidirected graph whose nodes are the balance equations and whose edges are the arcs (each arc lies on a side of each of the two patches it borders). When this graph is connected and has a cycle that ± 1 values cannot balance, as on the body, the companion paper shows that a set of free arcs parametrizes the lattice of whole balanced lengths: here $2{,}069 - 1{,}552 = 517$ arcs can be given any whole lengths and the others follow, so balances alone never obstruct whole lengths. Closing exactly, in whole numbers, the turns found closable together is feasible: HiGHS solves it in about two minutes a way for three of the four ways tried (the fourth, with 87 turns, did not finish within ten minutes). The solutions leave 1 to 4 held turns shifted where the engine leaves

way	held	kept by the projection	closable together
0	48	42	47
1	78	46	73
2	52	46	51
3	86	47	82
4	84	49	83
5	160	48	154
6	88	53	87
7	169	49	160

Table 2: Male body at 10,590 quads, real numbers, engine of the export. "Kept by the projection": by the clamped projection of that engine; "closable together": the largest set of held turns closable with every bound of separation kept.

targets of the integer search	turns closed in reals	still shifted after	spiral edges (mesh)
projection, clamped	about 50	10–23 of them	1.1%
linear program, L^1	up to 160	—	11.9%
active set, floor 0	all held (up to 169)	117 of 169	3.3%
active set, floor half the real length	36–72	—	9.8%

Table 3: Male body at 10,590 quads, engine of the export, one run each; the mesh keeps the way with the fewest spirals, so its share is that of a single way. The engine at the cited commit uses the active set with floor 0 together with the constraint rounds and the floor of the deviation cap, which this table does not include.

13 to 44, and fewer of the turns not held as well (5 against 9, 5 against 11, 0 against 27), at a deviation from the real lengths 4% to 62% higher. Yet the quad meshes they make spiral *more*: 12.2% of their edges against 8.8% for the first way, 3.7% against 1.1% for the second, and the third breaks the structure. Most of the loops that spiral are not turns the sampled walks find.

Walking every strip of the mesh of way 2 instead (Proposition 2) finds 230 ways round, each strip twice, once in each direction; 168 come back shifted. Selecting them by lean, as the constraint rounds do, keeps 222, of which 218 (109 strips) can be closed together in real numbers with every bound kept. Posed directly to HiGHS, they gave no whole solution in ten or thirty minutes; given the 171 arcs that every real solution sets to zero (each certified by a non-negative combination of shifts, 23 distinct certificates round fingers, toes, feet, head, torso and neck), it finds one in a second. The obstruction is the size: no real lengths closing the 218 rings have a mean deviation from the targets below 0.817, where the engine of the export refused 0.474 (1.5 times the 0.316 of its plain search; the engine at the cited commit allows 0.516). A subset of 190, chosen greedily, closes at 0.399 in real numbers, keeping the number of quads to first order. In whole numbers, with the reformulation of Aardal, Hurkens and Lenstra [1] on an LLL-reduced basis [21] of the lattice of closed quantizations, HiGHS, stopped after 1,200 seconds, finds whole lengths closing all 190 within every bound at a mean deviation of 0.619 and 10,568 quads, and proves that none does better than 0.536. Given to the engine in place of its lengths, they close the 190 rings but make 2.5% of spiral edges against 1.1%: at the new lengths the strips are other walks, 156 of the 230 ways round shifted. Rounding costs between 0.137 (proven) and 0.220 (found) of mean deviation above the real optimum; it is the choice of the turns to close, at the lengths where they are the strips of the mesh, that decides how much the mesh spirals.

The heuristic leaves most of the gap. The companion paper [12] computes the least number of held turns that any lengths in P_{eng} leave shifted, a minimum cover of the irreducible infeasible subsystems [26]: exactly 1, 5, 1, 4, 1, 6, 1 and 9 in ways 0 to 7, against 13, 39, 16,

44, 49, 115, 44 and 123 left shifted by the engine. With the engine's cap on the deviation and its window of sizes (10,379 to 10,801 quads) added, at whole lengths, the least number is still exactly one in ways 0 and 2: whole lengths closing every held turn but the wrist exist at 10,439 and 10,436 quads, at mean deviations 0.442 and 0.469. Of the 16 turns of way 2 the engine leaves shifted, the bounds of separation force one and the cap, the size and integrality none; the heuristic accounts for 15. Closing more turns, however, is not what the measure of spirals sees: a local search over whole lengths within the same cap and size closes 38 more strips of the engine's mesh than the engine does, but at its lengths, which also break the mesh, the strips are again other walks, and the share of spiral strips is 0.020 against 0.011 for the engine. Choosing the most turns to close is NP-hard in general, already for the bidirected balances and 0–1 bounds of quantization [12].

Closing the most rings, end to end. We measured what an exact selection buys on the bench, with the engine of branch `complete-helices` (the constraint rounds without the finer work mesh at thin walls of the cited commit: 4.54% of spiral edges over the 36 runs). For the lengths the engine keeps, a mixed-integer program closes as many of its strips leaning less than 0.12 as it can, weighted by their length round, within the balances, the bounds of separation, the cap and the quad budget; the engine meshes the result, the strips are walked again at the new lengths, and the program solved again, three rounds. With the shift expressions alone the program closes 27.3 of the 38.2 such strips of a model (mean over the 36 runs), but at the new lengths only 22.1 are closed strips, against 21.8 for the engine. Required for every strip counted as closed, the corridor inequalities (1) make the count exact (24.6 closed by the program, 24.0 found closed by walking the mesh). In place of the engine's lengths, the lengths found do not spiral less: 4.77% and 5.08% of spiral edges in two runs against 4.54%. Added to the candidates the engine chooses among by its own count of spirals, they lower the mean to 2.32% and 2.70%, never higher; three other draws of the engine's per-arc perturbation, added the same way, give 4.12%, three lengths drawn at random within the same cap and budget 4.34%, and the rounds without the corridor inequalities 3.15%, better than the other draws on 7 runs and worse on 7. Closed loops (38.0% against 36.3% of the edges) and mean distance (0.383 against 0.385) barely move, and all 36 meshes are valid. On the male body every round spiralled more than the engine (3.45% to 5.59% against 1.59%). The gain needs a solver that finishes: with a limit of 5 seconds per solve it vanishes (4.10% against 4.12% for the other draws), as 63 of the 126 pieces get no solution in time and 68 of 108 rounds return the engine's own lengths. A local search inside the engine, walks of its strip network biased by Lagrange multipliers with the corridor inequalities kept, costs 0.1 to 0.25 seconds per model but closes no more rings than the engine: a single walk almost always shifts a ring already closed, and closing needs combinations of walks.

Acting before quantization. Since the shift of a ring bounding a disc is, up to sign, the across component of the Burgers vector of the singular vertices inside it [12], moving the singular vertices in the parametrization is a natural lever. On seven closed models of genus zero (blob, rock, creature, spot, pig, ogre, blub), the Burgers sum reproduces the shift of 3,600 of the 3,604 rings of the seamless map that bound a disc and lean by at most 0.12, to $10^{-5}(1 + L)$ for a ring of length L. Yet moving a singular vertex barely moves the shift: the other cone points move in the map and absorb the change, and over twelve moves per model the measured change is 0.12, 0.03 and 0.01 of the predicted one on the blob, the creature and the pig. Rotating the field rigidly on the smaller side of a ring does move it (26% to 87% of the rotation times the length, as the rotated region grows by 0 to 40 rings of vertices), at almost no cost in distortion; but rotating the field cylinder by cylinder to bring the shifts to zero moved the twist onto rings of 90 to 200 edges, and over 36 runs 2 improved and 4 regressed, which we read as chaos rather than progress. We read this, as a conjecture, as saying that with a map of fixed scale the lever is the orientation of the field along whole cylinders, and that the conflicts are better attacked in

quantization.

8 Limitations and open problems

The results come from one implementation and are preliminary. The spiral share of a single model is chaotic, so only the means over three budgets are robust; the other figures are single runs. The measure sees helices of pitch up to four only. The comparison with other remeshers and the experiments of Section 7 used earlier engines (Appendix A), and the male body was not remeshed with the cited commit. The exact look covers one body, with an engine without the constraint rounds, and holds relative to the engine's bounds of separation, which are its choice of arcs and neither a subset nor a superset of the valid quantizations; the cap imposed there is the engine's only approximately (a quantization at 0.4737 was refused by it). Times were taken on a shared machine and the outcomes that depend on a time limit depend on its load. The scripts and logs of the exact experiments are not yet published (Appendix A).

Three problems remain open. *An exact formulation*: minimizing the deviation subject to the balances, the bounds and $s_\gamma(\ell) = 0$ for a set of turns is a minimum-deviation flow with linear side constraints; how much of the exact machinery of [13] carries over is open, and the linear relaxation already closes almost all held turns. *Closing in whole numbers*: whole lengths closing every ring asked for are found through a reduced lattice basis, but in minutes, with the number of quads held only to first order and 0.14 to 0.22 above the real optimum in deviation; can the integer search move within this lattice at the speed of a flow solver? *Which turns*, the central one on the evidence of Section 7: closing rings changes the lengths, the lengths change the strips, and the strips of the new mesh are other walks. Is there a finite set of turns whose closing implies that every loop of the quad mesh closes? Can the loops of the quad mesh be read off the T-mesh and its lengths directly, so that every one is accounted for? Which objective, short of all loops, predicts the spirals the measure sees?

AI assistance

This note and the software it describes were made by a human author working with an AI system, which is not an author. The author chose the problem and its aims (edge loops that close into rings and, as an artist, few singular vertices and few spirals rather than perfectly square quads, quads only), set the direction and the order of the work, decided which results to pursue and which changes to merge, and judged the results through the measurements, pictures and drafts presented to him. Claude Opus 5.5 (Anthropic), used through Claude Code in October 2026 with several sub-agents in parallel, wrote the changes to the engine and the experiment scripts, ran all the experiments and exact computations, searched the literature, stated and proved the propositions, and drafted the text. References were checked for existence, and the note of each bibliography entry says how much of the work was read. Weaker forms of Propositions 1 and 2 are checked in Lean 4 [10] in the companion paper; everything else here not checked by machine was written with AI help and checked by AI agents rereading it and testing it on the data, not by a human expert. The note has not been peer reviewed; the author, who publishes it, is responsible for its content.

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A Provenance of the figures

The engine changed while the experiments were made. The table says, for each group of figures, which version produced it and where its raw output is. “Commit message”: the figures are stated in the message of that commit of the repository; “working logs”: the output files and scripts of the experiment, kept by the author outside the repository and not yet published.

figures	engine	source
twelve models, three budgets, 4.65% (Section 6)	the cited commit	working logs of the protocol below; a second run of the same engine gave the same figures for the 36 meshes
Table 1	commits `7dc1727`, `c86740f`, `cf4a82c`	commit messages (also `ab8d89c`)
against other remeshers	commits `83f0967`, `4e9620c`	`bench/results/results.json` at those commits
hand-made male body	—	`bench/results/RESULTS.md`
male body, openclare	`ab8d89c`; `b0d37fa` to `8fd1932`	commit messages; closed loops, distance and corners: working logs
Figures 3 and 4, their numbers	commit `89232b0`, one run per mesh; QuadriFlow through Blender, on the source scaled by 100 and scaled back	`figures/README.md` (commands, numbers, source checksum) and `figures/make_figures.sh`, which remakes them
Section 7, up to "The heuristic leaves most of the gap"	branch `rd-exact` (export hook), engine before `3b3122d`	working logs, scripts and the exported systems; some summary figures also in commit messages (`6374097`, `38b9fb3`)
closing the most rings, end to end	branch `maxclose-rd`, engine of branch `complete-helices`	working logs; some figures only in the experiment's summary
acting before quantization	a patch of the engine of commit `83f0967`	working logs; the ratios over twelve moves, the 36 runs and the 3,600 of 3,604 rings only in the experiment's summary

The protocol of the twelve models. For each model m (creature, blob, suzanne, spot, pig, ogre, bob, jerry, blub, rock with the preset `organic`; cage and part with `hard-surface`) and each budget $Q \in \{2{,}970, 3{,}000, 3{,}030\}$: `openclare remesh bench/out/`$m$`/source.obj` *out.obj* `-preset` P `-quads` Q, then `openclare measure` on the source and the output with `-target` Q `-json`; the figures are the means of `loop_edges_spiral`, `loop_edges_closed`, `loops_spiral`, `mean_distance` and `angle_deviation_mean` over the 36 runs.

Reproducibility. The method is implemented in openclare (Rust), in the main branch of the repository at the cited commit; the bench is driven by `bench/run.py`. The export of the T-mesh used in Section 7 is an experimental hook of the branch `rd-exact`, and the end-to-end selection is on the experimental branch `maxclose-rd`; neither is part of the engine, and neither is tagged yet. Until the scripts and logs of the exact experiments are published, the measured statements of Section 7 can be checked against these branches but not rerun as they stand. This version of the note was built from commit `9a2851905a25`.